User Cases :

UC Name	**UC-01 Register**
Summary	This use case describes the process where a new user creates an account or an existing user logs into the system to access application features.
Dependency	None.
Actors	Customer (Primary Actor)
Preconditions	The user must have access to the application and internet.The user does not have an existing account.
Description of the main sequence	• The user navigates to the registration page. • The system displays the registration form. • The user enters required details (name, email, password). • The user submits the form. • The system validates the input. • The system creates a new account.
Description of the Alternative Sequence	• Invalid input: a. The system highlights errors. • Email already exists: a. The system prompts the user to use a different email.
Non functional Requirements	• The registration process should be completed within a few seconds. • Input validation should occur in real time. • User data must be securely stored (encrypted). • The interface must be simple and user-friendly.
Postconditions	A new user account is created successfully.

UC Name	**UC_02: Login**
Summary	This use case describes the process where an existing user logs into the system to access application features.
Dependency	UC_01 (Register)
Actors	Customer, Admin, Conductor
Preconditions	The user must have access to the application and internet. For login, the user must already have an existing account.
Description of the main sequence	• The user navigates to the login page. • The user enters credentials (email and password). • The system validates credentials. • The system grants access to the dashboard.
Description of the Alternative Sequence	Invalid credentials: a. The system displays an error message.
Non functional Requirements	• Authentication must be secure (encrypted passwords). • Login response time should be under 2 seconds. • The system should prevent unauthorized access. • Multiple failed attempts should be handled securely.
Postconditions	A valid user account is stored in the system .The user is successfully logged into the system.

UC Name	**UC-03 Manages Profile**
Summary	Allows the customer to view and update their personal account information.
Dependency	UC_02 (Login)
Actors	Customer (Primary Actor)
Preconditions	The user must be logged into the system.
Description of the main sequence	• The user navigates to the profile page. • The system displays current user information. • The user updates profile details. • The user submits changes. • The system validates input. • The system saves updated data.
Description of the Alternative Sequence	Invalid input: a. The system displays validation errors.
Non functional Requirements	• Profile updates should be processed instantly. • Data validation must be accurate and immediate. • The interface should be easy to use. • User data must remain secure and consistent.
Postconditions	User profile is updated successfully.

UC Name	**UC-04 Search Routes**
Summary	Enables the customer to search for available routes based on destination and travel date.
Dependency	UC-02
Actors	Customer (Primary Actor)
Preconditions	User logged in accesses search functionality.
Description of the main sequence	• The user enters origin, destination, and date. • The system processes the query. • The system displays matching routes.
Description of the Alternative Sequence	No routes found → message shown.
Non functional Requirements	• Search results should be displayed quickly (within 2 seconds). • The system should handle multiple search requests efficiently. • The interface should support easy filtering and navigation.
Postconditions	A list of available routes is displayed.

UC Name	**UC-05 View Schedules**
Summary	This use case allows allows the customer to view available bus schedules and filter them based on preferences.
Dependency	UC-02
Actors	Customer (Primary Actor)
Preconditions	The user is logged in.
Description of the main sequence	• The user navigates to the schedules section. • The system displays available schedules. • The user filters or searches schedules. **Includes:** UC_04 (Search Routes)
Description of the Alternative Sequence	No schedules available → system informs user
Non functional Requirements	• Schedule data must be accurate and up-to-date • System must load schedule details instantly • UI must present data clearly
Postconditions	User is informed about available schedules.

UC Name	**UC-06 Select Seat**
Summary	This use allows the customer to choose a preferred seat during the ticket booking process.
Dependency	None
Actors	Customer (Primary Actor)
Preconditions	User should be logged in.
Description of the main sequence	• System displays seat layout. • User selects seat. • System reserves seat temporarily.
Description of the Alternative Sequence	Seat unavailable-> Try selecting another seat.
Non functional Requirements	• Seat availability must update in real time. • The system should prevent double booking. • The seat layout must be visually clear and responsive.
Postconditions	Seat selected.

UC Name	**UC-07 Make Payment**
Summary	Handles the payment process for ticket purchase, ensuring secure transaction completion.
Dependency	UC-05
Actors	Customer (Primary Actor)
Preconditions	Seat is selected.
Description of the main sequence	• User enters payment details. • System processes payment. • Payment confirmed
Description of the Alternative Sequence	Payment fails ,user should try again or log out.
Non functional Requirements	• Payment transactions must be secure (HTTPS, encryption). • Payment processing should be fast and reliable. • The system must ensure no data loss during transactions. • Compliance with payment standards is required.
Postconditions	User has completed the transaction.

UC Name	**UC-08 Generate Ticket**
Summary	Generates a digital ticket after successful payment, including all travel details.
Dependency	UC-07
Actors	System
Preconditions	Payment successful.
Description of the main sequence	Payment successful.
Description of the Alternative Sequence	If there are no bookings system displays empty state.
Non functional Requirements	• Ticket generation should be completed instantly after payment. • The system must ensure accuracy of ticket data. • The ticket format should be standardized and readable.
Postconditions	Ticked is generated.

UC Name	**UC-09 Purchase Ticket**
Summary	This use case represents the complete process of booking a ticket, including seat selection, payment, and ticket generation.
Dependency	UC_06, UC_07, UC_08
Actors	Customer (Primary Actor)
Preconditions	• User is logged in. • Route selected.
Description of the main sequence	• User selects route. • User selects seat. • User makes payment. • System generates ticket. • Confirmation shown.
Description of the Alternative Sequence	Payment failure → system retries or notifies user
Non functional Requirements	• The booking process should be smooth and uninterrupted. • The system must maintain consistency across all steps. • High reliability is required during the full transaction flow.
Postconditions	User has purchased the ticket.

UC Name	**UC-10 Download Ticket**
Summary	Allows the customer to download the purchased ticket in a digital format (e.g., PDF).
Dependency	UC_09 Purchase Ticket
Actors	Customer (Primary Actor)
Preconditions	Ticket exists and is generated.
Description of the main sequence	• User selects ticket. • Clicks download. • System downloads file.
Description of the Alternative Sequence	Delivery failure → system retries or notifies user
Non functional Requirements	• Download should start immediately upon request. • The file should be accessible on all devices. • The system should ensure file integrity.
Postconditions	Ticket saved.

UC Name	**UC_11: View Bookings**
Summary	Enables the customer to view all current and past bookings.
Dependency	UC_02 Login
Actors	Customer
Preconditions	User is logged in.
Description of the main sequence	• Open bookings. • System shows bookings.
Description of the Alternative Sequence	Delivery failure → system retries or notifies user
Non functional Requirements	• Booking history should load quickly. • Data must be accurate and up-to-date. • The interface should be clear and easy to navigate.
Postconditions	Booking list displayed.

UC Name	**UC_12: Manage Routes**
Summary	Allows the admin to create, update, and delete bus routes in the system.
Dependency	UC_02 Login
Actors	Admin
Preconditions	• The admin is logged into the system. • The system database is available.
Description of the main sequence	• The admin navigates to the route management section. • The system displays the list of existing routes. • The admin chooses to add, edit, or delete a route. • The admin enters or modifies route details (origin, destination, distance, duration). • The admin submits the changes. • The system validates the input data. • The system saves the updated route information. • The system displays a confirmation message.
Description of the Alternative Sequence	• Invalid input: a. The system highlights incorrect fields. b. The admin corrects the data. • System error: a. The system displays an error message and does not save changes.
Non functional Requirements	• The system should provide fast response time when loading routes. • The interface should be clear and easy to use. • Data integrity must be ensured during updates.
Postconditions	• Route information is successfully updated in the system. • Changes are reflected immediately in the database.

UC Name	**UC_13: Manage Schedules**
Summary	Enables the admin to manage travel schedules for different routes.
Dependency	UC_02 Login
Actors	Admin
Preconditions	• The admin is logged in. • Routes already exist in the system.
Description of the main sequence	• The admin accesses the schedule management section. • The system displays existing schedules. • The admin selects to add, update, or delete a schedule. • The admin enters schedule details (route, departure time, arrival time, date). • The admin submits the data. • The system validates the information. • The system saves the schedule. • The system confirms the operation.
Description of the Alternative Sequence	• Invalid schedule data: a. The system shows validation errors. • Conflict in schedule (e.g., overlapping times): a. The system alerts the admin.
Non functional Requirements	• The system should prevent scheduling conflicts. • High system reliability is required. • Data should be updated in real time.
Postconditions	Schedule data is stored and updated correctly.

UC Name	**UC_14:Manage Buses**
Summary	Allows the admin to add and update bus details such as capacity and type.
Dependency	UC_02 Login
Actors	Admin
Preconditions	Admin is logged in.
Description of the main sequence	• The admin navigates to the bus management section. • The system displays existing buses. • The admin adds, edits, or deletes bus information. • The admin enters details (bus ID, capacity, type). • The admin submits the changes. • The system validates the input. • The system saves the data. • The system confirms the update.
Description of the Alternative Sequence	• Invalid input: a. The system displays error messages.

	• Duplicate bus ID: a. The system rejects the entry.
Non functional Requirements	• The system must ensure data accuracy. • Fast data retrieval and updates.
Postconditions	Bus information is updated successfully.

UC Name	**UC_15: Monitor Seat Availability**
Summary	Enables the admin to view seat availability across routes and schedules in real time.
Dependency	UC_02 Login
Actors	Admin
Preconditions	• The admin is logged in. • Active schedules exist.
Description of the main sequence	• The admin accesses seat availability monitoring. • The system displays seat status for each bus and route. • The admin reviews available and occupied seats. • The system updates data in real time.
Description of the Alternative Sequence	No available data: a. The system displays a message indicating no schedules available.
Non functional Requirements	• Real-time updates are required. • High performance for large datasets.
Postconditions	The admin has an updated view of seat availability.

UC Name	**UC_16: Monitor Ticket Inventory**
Summary	Allows the admin to track available, sold, and remaining tickets.
Dependency	UC_02 Login
Actors	Admin
Preconditions	• The admin is logged in. • Ticket data exists.
Description of the main sequence	• The admin opens the ticket inventory section. • The system retrieves ticket data. • The system displays total tickets, sold tickets, and remaining tickets. • The admin reviews inventory details.
Description of the Alternative Sequence	System cannot retrieve data: a. The system displays an error message.
Non functional Requirements	• Data should be accurate and up-to-date. • Fast system response time.
Postconditions	The admin has a clear overview of ticket inventory.

UC Name	**UC_17: Track Ticket Sales**
Summary	Provides the admin with insights and statistics about ticket sales over time.
Dependency	UC_02 Login
Actors	Admin
Preconditions	• The admin is logged in. • Ticket sales data exists.
Description of the main sequence	• The admin navigates to ticket sales tracking. • The system retrieves sales data. • The system displays statistics (daily, weekly, monthly sales). • The admin analyzes performance.
Description of the Alternative Sequence	No sales data available: a. The system displays a notification.
Non functional Requirements	• Data visualization should be clear and understandable. • The system should process large datasets efficiently.
Postconditions	The admin gains insights into ticket sales performance.

UC Name	**UC_18: Generate Reports**
Summary	Allows the admin to generate detailed reports based on system data and sales.
Dependency	UC_17 (Track Ticket Sales)
Actors	Admin
Preconditions	• The admin is logged in. • Sales and system data are available.
Description of the main sequence	• The admin selects report criteria (date range, routes, buses). • The admin requests report generation. • The system processes the data. • The system generates the report. • The system displays the report. • The admin may export the report (PDF/Excel).
Description of the Alternative Sequence	• Invalid criteria: a. The system requests correction. • System error: a. The system displays an error message.
Non functional Requirements	• Reports should be generated quickly. • The system must ensure data accuracy. • Export functionality should be reliable.
Postconditions	Report is successfully generated and available for viewing/export.

UC Name	UC_19: View Passenger List
Summary	Enables the conductor to view the list of passengers for a specific trip.
Dependency	UC_02 (Login)
Actors	Conductor
Preconditions	• The conductor is logged into the system. • There are scheduled trips with booked passengers.
Description of the main sequence	• The conductor navigates to the passenger list section. • The system retrieves data for the selected trip. • The system displays the list of passengers, including names, seat numbers, and ticket status. • The conductor reviews the passenger list before or during the trip.
Description of the Alternative Sequence	• No passengers found: a. The system displays a message indicating no passengers for the selected trip. • System error: a. The system shows an error message and prompts retry.
Non functional Requirements	• The list should load quickly even with many passengers. • The interface should be clear and easy to read. • Data must be accurate and up-to-date.
Postconditions	The conductor has access to the complete passenger list for the trip.

UC Name	UC_20: Check Ticket Details
Summary	Allows the conductor to verify ticket information such as passenger name, seat, and route.
Dependency	UC-08 Generate Ticket
Actors	Conductor
Preconditions	• The conductor is logged in. • A ticket has been scanned or selected.
Description of the main sequence	• The system retrieves ticket information. • The system displays details such as passenger name, route, seat number, and ticket status. • The conductor verifies the ticket information.
Description of the Alternative Sequence	• Ticket not found: a. The system displays an invalid ticket message. • Data retrieval error: a. The system shows an error notification.
Non functional Requirements	• Fast retrieval of ticket data. • High data accuracy.
Postconditions	The conductor verifies whether the ticket is valid.

UC Name	**UC_21: Mark Ticket as Used**
Summary	Updates the ticket status to “used” after validation during boarding.
Dependency	UC-10 Download Ticket
Actors	Conductor
Preconditions	• The conductor is logged in. • The ticket is valid and not previously used.
Description of the main sequence	• The conductor confirms ticket validation. • The system updates the ticket status to “Used”. • The system saves the updated status. • The system confirms the action.
Description of the Alternative Sequence	• Ticket already used: a. The system displays a warning message. • Invalid ticket: a. The system rejects the action.
Non functional Requirements	• Real-time status update. • High system reliability.
Postconditions	The ticket is marked as used in the system.

UC Name	**UC_22: Scan Ticket**
Summary	Allows the conductor to scan and validate tickets using a QR code or barcode
Dependency	• UC_20 (Check Ticket Details) • UC_21 (Mark Ticket as Used)
Actors	Conductor
Preconditions	• The conductor is logged into the system. • The conductor has access to a scanning device (QR/barcode scanner)
Description of the main sequence	• The conductor opens the ticket scanning feature. • The conductor scans the ticket (QR code or barcode). • The system reads the ticket data. • The system validates the ticket. • The system displays ticket details. • The conductor confirms validity. • The system marks the ticket as used.
Description of the Alternative Sequence	• Invalid ticket: a. The system displays an error message. • Ticket already used: a. The system shows a warning and prevents reuse. • Scan failure: a. The system prompts the conductor to rescan.
Non functional Requirements	• Fast and accurate scanning process. • Real-time validation. • High system availability during trips.
Postconditions	• The ticket is validated and updated in the system. • The system prevents duplicate usage of the same ticket.